Beyond Semantic Confidence: Relation-Algebra-Constrained Geometric Compatibility for 3D Scene Graph Relations

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Abstract

3D Scene Graph predictors can assign high confidence to relation candidates whose predicates are inconsistent with reconstructed object-pair geometry. We formulate this as a post-source compatibility problem and introduce RelCompat3D, which separates predicate semantics, same-pair geometry, and source confidence. Its predictor-agnostic compatibility model is trained without source scores using linked positive–counterfactual ordering and is projected onto the applicable proximity and vertical relation-algebra orbits. Source confidence is introduced only during re-ranking. We evaluate exact-label Recall@K jointly with verifier-derived Violation@K across VL-SAT, Open3DSG, and SGFN on a shared 3DSSG target. At $K = 100$, RelCompat3D improves Open3DSG Recall from 0.5161 to 0.6055 while reducing Violation from 0.1242 to 0.0339, with analogous aggregate gains on VL-SAT and SGFN. Strong fusion baselines, factor controls, uncertainty sensitivity, and family-wise analysis show that the gains transfer across predictors while contact-heavy relations remain challenging.

Introduction

3D Scene Graphs turn reconstructed scenes into structured object-relation representations that can support spatial reasoning, embodied AI, scene alignment, visual grounding, and downstream decision making (Wald et al. 2020; Sarkar et al. 2023; Xie, Pagani, and Stricker 2024; Liu et al. 2026). For these uses, relation prediction must be supported by the reconstructed scene, not only plausible at the category or language level. A graph edge such as `standing on`, `next to`, or `higher than` is useful only if it describes the actual subject-object pair in 3D.

We study a failure mode in which relation predictors rank plausible labels that are inconsistent under explicit tests of the actual object-pair geometry. A relation can sound plausible for two object categories while being contradicted by their contact evidence, distance, or vertical arrangement in the reconstructed scene. This mismatch matters more as 3D Scene Graphs become interfaces for language-facing, open-vocabulary, and embodied systems (Koch et al. 2024b; Hou et al. 2025; Zhang et al. 2025; Wang et al. 2025; Saxena and Chiun 2025; Yu et al. 2025; Madhavaram et al. 2026).

The failure is not simply that existing 3DSSG methods ignore geometry. Prior work already uses edge features, point-cloud structure, spatial knowledge, multimodal semantics, and geometric fusion (Zhang et al. 2021; Feng et al. 2023; Wang et al. 2023; Koch et al. 2024a; Chen et al. 2024). The gap is more specific: a source relation score—even when its predictor already uses geometry—is not necessarily an explicit estimate of whether the same subject-object pair satisfies the predicate in 3D. This makes the problem an AI reliability issue, not only a 3DSSG implementation detail: language-facing and embodied systems can inherit a confident symbolic relation that contradicts the underlying world state.

This failure suggests a factor-isolated design. RelCompat3D preserves each relation candidate's subject–object identity, joins its reconstructed geometry, and separates predicate semantics T, same-pair geometry G, and source score Z. A predictor-agnostic compatibility model excludes Z and predictor identity; Z enters only during re-ranking. Linked counterfactuals teach the model which geometry supports a predicate, while orbit projection enforces proximity symmetry and vertical inverse equivariance exactly. The resulting structured product retains source-score magnitudes, and scale-robust fusion, family-conditioning, and rule-filtering comparisons isolate its behavior. Figure 1 summarizes the framework.

We evaluate support/contact, proximity, and relative-vertical relations because their constraints are identifiable from reconstructed ordered-pair geometry. Open3DSG, VL-SAT, and SGFN provide complementary open-vocabulary and closed-set predictors (Wang et al. 2023; Koch et al. 2024b). The evaluation reports exact-label recall together with verifier-derived violation, uncertainty, coverage, paired confidence intervals, and family composition. We do not extrapolate this cross-predictor evidence across dataset ontologies.

Our contributions are threefold:

1. We formalize the mismatch between relation confidence and instance-level geometric consistency through identity-preserving evaluation and joint exact-label Recall@K–verifier-derived Violation@K.
2. We introduce a source-score-excluded compatibility

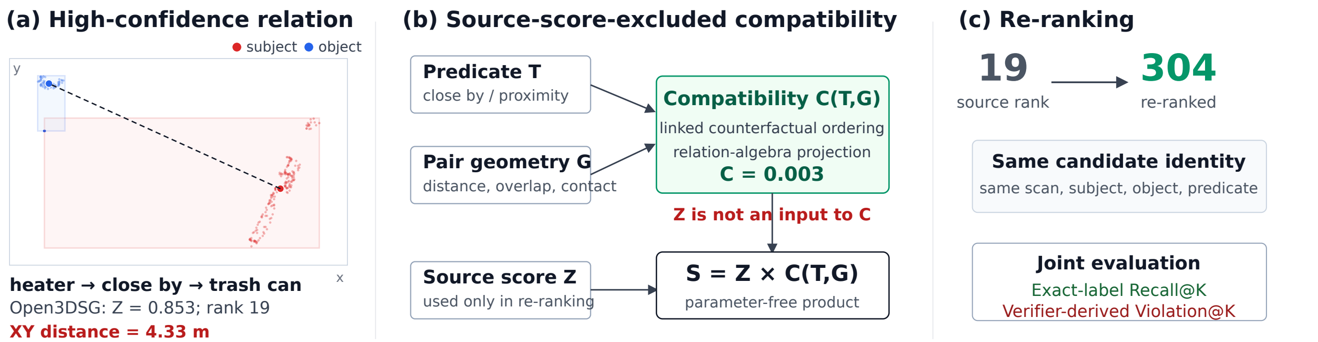

Figure 1: **Failure mechanism and RelCompat3D overview.** A high-scoring Open3DSG relation is contradicted by the actual ordered-pair geometry. RelCompat3D preserves the relation-row identity, isolates predicate semantics T, same-pair geometry G, and source confidence Z, and estimates $C^{\mathrm{alg}}(T, G)$ without Z. In the shown case, low compatibility moves the relation from rank 19 to 304. Linked-counterfactual and relation-algebra controls test the factorization; evaluation jointly reports exact-label recall and verifier-derived violation.

model that combines linked counterfactual ordering with exact projection onto the applicable relation-algebra orbit.

3. We apply one unchanged compatibility model to VL-SAT, Open3DSG, and SGFN, evaluating cross-predictor robustness with strong fusion baselines, factor controls, uncertainty sensitivity, and family decomposition.

The structured product improves the aggregate $K = 100$ recall–violation operating point for all three predictors, while smaller budgets expose predictor-dependent trade-offs. The result is a reliability layer for geometry-checkable relations rather than a replacement relation generator.

Related Work

3D Scene Graph relation prediction. 3D Scene Graphs organize indoor scenes as object-relation structures grounded in 3D space (Armeni et al. 2019). The 3RScan/3DSSG setting established predicate recall for 3D relation prediction (Wald et al. 2020), while incremental and online systems study how scene graphs can be constructed from RGB-D streams or spatial perception stacks (Wu et al. 2021; Hughes, Chang, and Carlone 2022). Recent predictors improve closed-set predicates using edge reasoning, point-cloud features, multi-modal cues, spatial knowledge, self-supervised pretraining, object-centric features, and edge-centric relational reasoning (Zhang et al. 2021; Feng et al. 2023; Wang et al. 2023; Koch et al. 2024a; Heo et al. 2025; Ma et al. 2026). These methods already use geometry; RelCompat3D targets the complementary post-source question of whether a high-scoring predicate is geometrically compatible with the same reconstructed object pair.

Open-vocabulary 3D graphs. Open-vocabulary 3D perception provides queryable object and region features (Peng et al. 2023; Takmaz et al. 2023); graph systems then connect such features to compact scene representations for querying, planning, navigation, online mapping, and language interaction (Gu et al. 2024; Werby et al. 2024; Maggio et al. 2024). Recent 3D scene graph work extends this line toward open-vocabulary objects, open-set relations, VLM features, online graph generation, and functional relations (Koch et al. 2024b; Chen et al. 2024; Hou et al. 2025; Zhang et al. 2025; Wang et al. 2025). This line raises the reliability requirement: when relation edges are language-facing scene representations, plausible text labels must still refer to object pairs that satisfy the corresponding spatial relation.

Geometry-aware relation evidence. A broad set of 3D reasoning methods already uses geometry: edge features, support/contact patterns, affordance geometry, metric distance, topology, and semantic-geometric fusion (Zhang et al. 2021; Feng et al. 2023; Huang et al. 2025; Shao et al. 2025; Xie, Pagani, and Stricker 2024). Scene graph alignment and registration methods further demonstrate that graph structure and geometric consistency matter when scene graphs are reused for matching, alignment, and downstream geometric tasks (Sarkar et al. 2023; Xie, Pagani, and Stricker 2024).

RelWitness uses visual-geometric witnesses, calibrated witness quality, and witness-consistent decoding for open-vocabulary 3D scene graph generation (Nguyen et al. 2026). Statistical confidence rescoring refines 3D scene graph predictions from multi-view images using semantic masks, neighboring relations, and statistical priors (Yeo, Li, and Lee 2025). These methods modify relation generation or learn source-specific rescoring. RelCompat3D instead operates on fixed candidate outputs and estimates same-pair compatibility with neither predictor identity nor source score as input.

Recent reliability-adjacent systems sharpen this boundary. SGFormer++ uses a spatial feature adapter inside an incremental predicate predictor (Qi et al. 2026); RelGraphOV prunes geometrically implausible relations while learning an open-vocabulary scene representation (Chen et al. 2026); and PUF performs uncertainty-aware semantic–spatial fusion for online node association and evidence accumulation (Yang et al. 2026). In contrast, RelCompat3D leaves each predictor fixed and re-ranks already-produced candidates using source-score-excluded, same-pair geometric compatibility.

Neau et al. mine spatial, functional, and relation-algebra constraints and use declarative reasoning to refine fixed image-SGG outputs, reporting a Constraint Violation Rate alongside task metrics (Neau et al. 2026). RelCompat3D differs in estimating continuous compatibility from reconstructed 3D geometry, preserving ordered-pair identity, and enforcing applicable 3D relation-algebra orbits before joint recall–violation evaluation.

Reliability evaluation. Standard R@K asks whether the correct predicate appears near the top, not whether the predicted relation is physically valid in the reconstructed scene. Confidence calibration asks whether model scores reflect empirical correctness likelihood (Guo et al. 2017), and selective prediction makes risk and coverage explicit (Geifman and El-Yaniv 2017). Inspired by this joint-utility view, RelCompat3D standardizes prediction rows, joins same-pair 3D evidence, and reports exact-label recall together with verifier-derived violation. Its compatibility output is a bounded ranking score, not a probability of physical validity.

Method

RelCompat3D is a post-source reliability layer rather than a replacement 3D Scene Graph generator. It preserves relation-candidate identity, estimates predicate-conditioned geometric compatibility without access to source confidence, and combines the two factors only during re-ranking. Figure 1 summarizes the pipeline.

Problem Setup and Reliability Objective

Let a relation source produce candidate rows

$$r_i = (\mathrm{scan}_i, \mathrm{context}_i, s_i, o_i, p_i, a_i, Z_i), \tag{1}$$

where s_i and o_i are ordered subject and object instance identifiers, p_i is the predicate, a_i its mapped relation family, and Z_i the source relation score. A context is the source-defined ranking unit associated with one scene subgraph; top-K selection is performed independently within each context. The key $(\mathrm{scan}_i, \mathrm{context}_i, s_i, o_i)$ preserves the same object pair across prediction, geometry, and ground-truth joins. We expose predicate/family semantics as $T_i = (p_i, a_i)$ and join raw pair geometry

$$G_i = G(\mathrm{scan}_i, \mathrm{context}_i, s_i, o_i). \tag{2}$$

The measured scope is

$$\mathcal{A} = \{\mathrm{support/contact},\ \mathrm{proximity},\ \ \mathrm{vertical\,order}\}, \tag{3}$$

whose constraints are identifiable from reconstructed ordered-pair geometry. Rows outside these families are excluded from the measured violation claim.

Our objective is to estimate a bounded compatibility score

$$C_i = \sigma(h_{a_i}(\Phi(T_i, G_i))) \in [0,1], \qquad Z_i \notin \mathrm{inputs}(C_i), \tag{4}$$

then enforce the applicable relation algebra to obtain C_i^{alg} and form a final ranking score $S_i = F(Z_i, C_i^{\mathrm{alg}})$. The score measures compatibility with a training target comprising ground-truth positives and counterfactual negatives; it is not a probability of physical validity. Predicate-aligned transformations are $T_i \times G_i$ interactions rather than raw geometry. Neither source score nor predictor identity enters C_i, so one model can be applied unchanged to different predictors on the shared target.

Let $L_K(c)$ be the top-K measured-family rows for context c and Y_c its exact-label ground-truth set. We jointly report

$$\begin{aligned} \mathrm{R@K} &= \frac{\sum_c |L_K(c) \cap Y_c|}{\sum_c |Y_c|}, \\ \text{Verifier-V@K} &= \frac{1}{|D_K|} \sum_{(c,i) \in D_K} \mathbf{1}[v_i = \mathrm{violated}]. \end{aligned} \tag{5}$$

where D_K contains selected rows with a supported verifier status $v_i \in \{\mathrm{satisfied}, \mathrm{uncertain}, \mathrm{violated}\}$. Family mapping never relaxes exact predicate matching. Because uncertain rows can lower the all-status ratio without being declared satisfied, the experiments also report decidable-only violation, uncertainty, pessimistic violation, and coverage.

Relation-Algebra-Constrained Compatibility

For each identity-preserving relation row, the raw geometry vector G contains pair distance and scale, centroid and extent differences, oriented-box overlap, and point-level contact/support evidence when observable. The feature map $\Phi = (\phi_T, \phi_G, \phi_\times)$ separates predicate/family indicators, predicate-independent geometry, and explicit predicate–geometry interactions. Source score, rank, source identity, and object-class shortcuts are excluded.

We first define a family logit and bounded compatibility score

$$\begin{aligned} h_a(\Phi(T_i, G_i)) &= w_a^\top \Phi(T_i, G_i), \\ C_i &= \sigma(h_a(\Phi(T_i, G_i)))\,. \end{aligned} \tag{6}$$

For proximity, τ swaps the ordered endpoints and leaves `close by` unchanged. For vertical order, τ swaps the endpoints and maps `higher than`↔`lower than`. These transforms encode

$$\begin{aligned} C_{\mathrm{close}}(s,o) &= C_{\mathrm{close}}(o,s), \\ C_{\mathrm{higher}}(s,o) &= C_{\mathrm{lower}}(o,s). \end{aligned} \tag{7}$$

Support/contact uses the singleton orbit because a family-wide endpoint swap is not valid.

Training uses two complementary structural signals. First, orbit-consistent augmentation preserves the predicate when proximity endpoints are swapped, whereas vertical order swaps endpoints and applies the inverse predicate. Second, for every linked positive–counterfactual pair $(i^+, i^-) \in \mathcal{P}$, the logits $\ell = w_a^\top \Phi$ receive a fixed margin penalty:

$$\begin{aligned} \mathcal{L}_a = \mathcal{L}_{\mathrm{BCE}} &+ 0.25\,\mathbb{E}_{\mathcal{P}}\Big[\log\Big(1 + e^{1-(\ell_{i^+} - \ell_{i^-})}\Big)\Big] \\ &+ 10^{-4}\|w_a\|_2^2. \end{aligned} \tag{8}$$

The 60,208 training rows yield 33,961 linked pairs; algebra augmentation gives 86,032 optimization rows. Numeric features are standardized and missing values imputed using training-only statistics. We use 800 deterministic full-batch steps at learning rate 0.2. The three family models contain 22–24 parameters.

Finally, we project the learned score onto the declared orbit:

$$C_i^{\mathrm{alg}} = \frac{1}{|\mathcal{O}_i|} \sum_{(T',G')\in\mathcal{O}_i} \sigma(h_{a_i}(\Phi(T',G'))) . \tag{9}$$

This makes proximity symmetry and vertical inverse equivariance exact at inference rather than merely encouraged by a penalty. The pairwise objective still distinguishes each linked positive from its counterfactual; projection does not consume source confidence.

Re-ranking and Controls

Our main score preserves source magnitudes through

$$S_i^{\mathrm{alg-prod}} = Z_i C_i^{\mathrm{alg}}. \tag{10}$$

For positive scores, $\log S_i = \log Z_i + \log C_i^{\mathrm{alg}}$; compatibility therefore contributes an additive log-score term without an additional fusion parameter. We also test scale-robust fusion. Let q_i^Z and q_i^C be descending within-context percentile ranks and r_i^Z, r_i^C their one-indexed ranks. Rank-average and fixed-constant Reciprocal Rank Fusion are

$$S_i^{\mathrm{rank}} = \tfrac{1}{2}(q_i^Z + q_i^C), \qquad S_i^{\mathrm{RRF}} = \frac{1}{60+r_i^Z} + \frac{1}{60+r_i^C}. \tag{11}$$

They provide scale-robust fusion comparisons.

Other comparisons isolate the mechanism. A pooled product replaces the three family models with one $T \times G$ compatibility model fitted on the training split. Compatibility-only removes Z, and an unprojected family product ablates relation-algebra projection. A 69-parameter nonlinear rescorer is matched to the total compatibility parameter count and trained with source-specific exact-label supervision.

Linked-counterfactual ordering and wrong-predicate tests ask whether the model uses the intended $T \times G$ interaction. Proximity-swap and vertical-inverse checks test Eq. 9; no blanket endpoint test is applied to support/contact. Identity-preserving joins and the T-only, raw-G-only, additive, shuffled-geometry, and mismatched-pair ablations test score leakage and geometry shortcuts. Together these comparisons test whether the gain follows from factor isolation and algebraic structure rather than an arbitrary fusion choice.

Experiments

Experimental Setup

Datasets and evaluation scope. Open3DSG, VL-SAT, and SGFN provide complementary open-vocabulary and closed-set relation predictors. Together they test cross-predictor robustness under one geometry-identifiable 3DSSG target. All use the same scope: 157 scans, 548 relation contexts, 36,808 directed object pairs, and 3,972 exact-label ground-truth relations from support/contact, proximity, and relative vertical families. Ground-truth membership is used only when computing recall, never to filter or rank candidate relations. The supplement evaluates relative size as a secondary scope extension; a fixed robust-point rule is equally strong within that family, so it is not used as core learned-method evidence.

Open3DSG uses the official non-averaged model. A documented preprocessing recovery completes the common evaluation scope; checkpoint identity and the recovery procedure are reported in the supplement.

Conditions and training boundary. We report the original source relation score, the relation-algebra-constrained product, rank-average fusion, pooled-compatibility product, and Reciprocal Rank Fusion. A 69-parameter nonlinear rescorer provides a parameter-matched, source-supervised comparison. Controls comprise wrong-predicate, compatibility-only, distance-only, shuffled-geometry, mismatched-pair, and label-fixed endpoint-swap rankings. Wrong-pair geometry is rotated within context and predicate; shuffled geometry is rotated across the source predicate stream. The endpoint control swaps proximity/vertical geometry while retaining the original predicate and leaves support/contact unchanged, for which no blanket swap is defined. The compatibility model uses only T, predicate-independent G, and explicit $T \times G$ interactions; source score and source identity are excluded. All normalization, imputation, feature preprocessing, and model parameters are estimated from the 1,061 training scans. The 117-scan development split is used for counterfactual discrimination, linked-pair ordering, and relation-algebra checks. The 157-scan official validation split contains no fitting, normalization, or imputation rows. The nonlinear comparison is the sole source-supervised exception: it is trained on exact-label correctness from the 117 development scans and evaluated without refitting on official validation. All cross-predictor statements concern the shared 3DSSG/3RScan target.

Metrics and uncertainty. Recall@K uses exact predicate labels within each relation context and is not directly interchangeable with Recall@K computed under a different candidate grouping. Violation@K is the fraction of selected, geometry-checkable rows classified as violated; uncertain rows are not labeled valid. We report the all-status definition, violation among decidable rows, uncertainty rate, pessimistic violation, and decidable/status coverage. We use $K \in \{5, 10, 20, 50, 100\}$. The primary endpoint is $K = 100$; all smaller budgets are reported without assigning additional selection roles. We report paired 95% bootstrap intervals over relation contexts; a scan-cluster sensitivity is provided in the supplement. Within-family and global-top-K family slices expose composition effects that may be hidden by the aggregate metric.

Cross-Predictor Recall–Violation Trade-off

Table 1 and Figure 2 show that the RelCompat3D product improves both metrics at $K = 100$ on all three predictors, with paired bootstrap intervals excluding zero for each Recall and V contrast. Open3DSG exposes the failure most strongly: the source ranks geometrically inconsistent edges at small K, whereas the structured product changes both recall and violation rather than only pruning edges. Near-ceiling VL-SAT leaves little recall headroom but retains a stable violation reduction. SGFN improves from $K = 10$ through $K = 100$; at $K = 5$ its recall is effectively unchanged and V increases

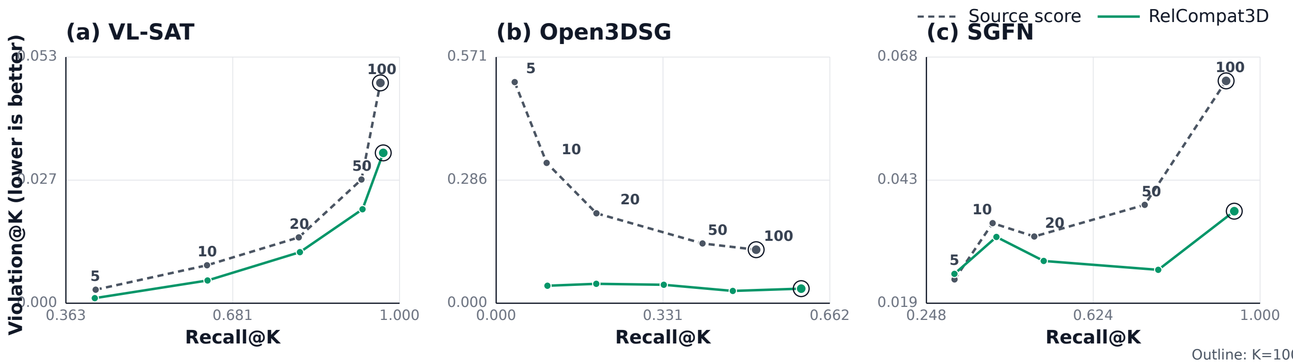

Figure 2: **Recall–violation trajectories.** Each line connects $K \in \{5, 10, 20, 50, 100\}$ for one predictor and ranking rule. Rightward and downward movement is preferred; outlined points denote $K = 100$.

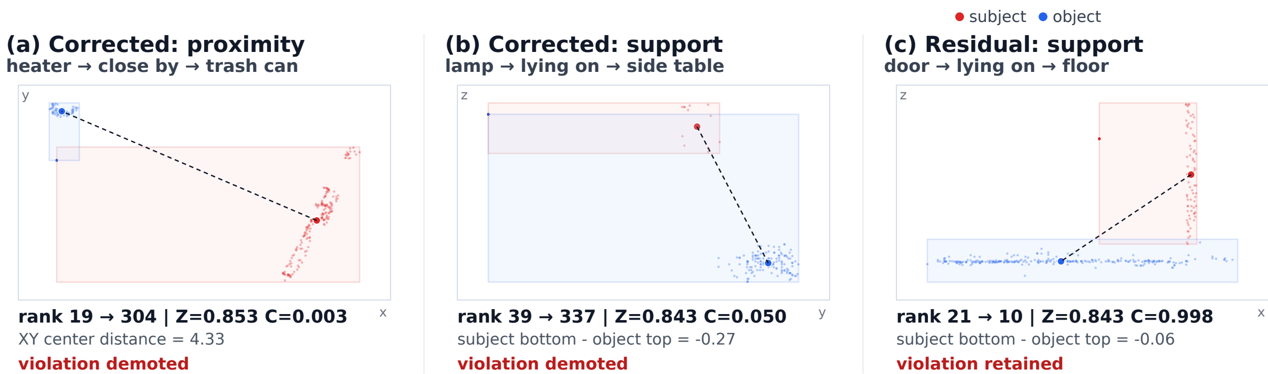

Figure 3: **Geometry-backed re-ranking cases.** Actual ordered-pair point clouds show successful proximity and support/contact demotions, plus a residual support error that remains at product rank 10 despite $C^{\text{alg}} \approx 1$. Red denotes the subject and blue the object.

slightly. The smaller-budget trade-off is therefore predictor dependent.

No comparator dominates the joint objective. Open3DSG's pooled product reaches the highest Recall@100 but more than doubles V relative to the RelCompat3D product. Rank-based fusion lowers V in several settings but incurs predictor-dependent Recall losses. These trade-offs favor the structured product as the predictor-agnostic operating point.

The scan-cluster sensitivity preserves the $K = 100$ directions when all contexts from a sampled scan move together. Recall intervals exclude zero for Open3DSG and SGFN and meet zero at the VL-SAT lower bound; every V interval remains below zero (supplement).

Structural and Factor Diagnostics

On the development split, linked positives outrank their counterfactuals in 99.23% of pairs, and correct vertical predicates outrank their inverses in 97.40% of cases. Projection makes proximity-swap and vertical-inverse errors exactly zero on development and across every evaluation row. These results establish the intended algebra and counterfactual ordering.

The unprojected product has nearly identical aggregate Recall, while projection removes the applicable algebra errors exactly. The factor ablation reports T-only, true raw-G-only, additive, and interaction conditions in the supplement. A parameter-matched source-supervised rescorer is slightly stronger on SGFN, showing the additional performance available from predictor-specific exact-label supervision.

Table 2 shows that the corruptions consistently break the joint operating point. Wrong-pair and shuffled geometry show that the gain depends on the identity-preserving join; wrong predicates and label-fixed swaps sharply increase V. Removing the source score can achieve low V but loses substantial Recall, confirming that geometry does not replace source confidence.

Uncertainty sensitivity gives the same aggregate conclusion: decidable-only V, uncertainty rate, and pessimistic V all decrease at $K = 100$ on every predictor, with paired intervals excluding zero. The all-status improvement therefore is not produced by moving failures into uncertainty.

Qualitative and Family Analysis

Figure 3 shows two successful demotions and one residual support error. The structured score moves the heater–trash-can proximity edge from rank 19 to 304 and the lamp–side-

Table 1: **Joint exact-label Recall (R) and verifier-derived Violation (V; lower is better).** Recall always uses the same 3,972-relation denominator; V uses selected rows with supported verifier status. The RelCompat3D product is the proposed score, rank-average is its scale-robust variant, RRF is a generic fusion comparator, and pooled product tests family conditioning.

		$K = 5$		$K = 10$		$K = 20$		$K = 50$		$K = 100$	
Source	Condition	R	V	R	V	R	V	R	V	R	V
Open3DSG	Source score	.0368	.5131	.1002	.3255	.1991	.2088	.4096	.1386	.5161	.1242
Open3DSG	RelCompat3D product	.1017	.0405	.1986	.0451	.3328	.0428	.4698	.0286	.6055	.0339
Open3DSG	Rank-average	.0997	.0219	.1815	.0308	.2928	.0354	.4718	.0394	.5994	.0531
Open3DSG	RRF	.0967	.0208	.1629	.0307	.2646	.0590	.4355	.0942	.5979	.0785
Open3DSG	Pooled product	.0929	.0522	.1765	.0584	.2963	.0544	.4718	.0528	.6443	.0747
VL-SAT	Source score	.4194	.0029	.6322	.0082	.8074	.0142	.9272	.0268	.9635	.0476
VL-SAT	RelCompat3D product	.4177	.0011	.6332	.0049	.8097	.0110	.9293	.0203	.9688	.0325
VL-SAT	Rank-average	.2110	.0113	.3517	.0120	.5287	.0168	.8119	.0191	.9617	.0248
VL-SAT	RRF	.2450	.0113	.4104	.0109	.6458	.0129	.8925	.0163	.9610	.0233
VL-SAT	Pooled product	.4172	.0011	.6312	.0062	.8112	.0117	.9300	.0219	.9690	.0387
SGFN	Source score	.3117	.0237	.3975	.0349	.4912	.0322	.7402	.0385	.9235	.0630
SGFN	RelCompat3D product	.3114	.0248	.4061	.0321	.5126	.0274	.7706	.0256	.9418	.0372
SGFN	Rank-average	.1707	.0252	.2870	.0219	.4421	.0203	.7314	.0203	.9474	.0268
SGFN	RRF	.1921	.0237	.3182	.0228	.4907	.0216	.7165	.0203	.9074	.0266
SGFN	Pooled product	.3142	.0237	.4079	.0325	.5159	.0278	.7925	.0292	.9413	.0464

table support edge from 39 to 337. The door–floor relation remains at rank 10 with $C^{\text{alg}} = 0.998$, illustrating that soft compatibility is not a validity label.

Family-wise intervals prevent aggregate overstatement. Relative vertical supplies most V reduction, proximity is unchanged at the within-family $K = 100$ ceiling, and support/contact V increases on all three predictors. The method improves aggregate selection but does not solve contact-heavy relations; full within-family and global-top-K composition tables are provided in the supplement.

Discussion and Limitations

One compatibility model transfers across three predictors on a shared, geometry-identifiable 3DSSG target, but this does not establish cross-dataset generalization or a universally optimal fusion rule. Rank fusion is stronger at some operating points and pooled compatibility can favor recall over violation. The contribution is predictor-agnostic compatibility and its joint reliability evaluation rather than formula superiority.

The aggregate effect is not family-uniform, and contact-heavy relations remain the main failure mode. Within-family and actual global-top-K slices are reported separately, so the aggregate result does not hide family-composition changes. More broadly, RelCompat3D assumes a geometry-identifiable predicate and known object instances. Functional and reference-frame-dependent predicates remain outside the evaluated scope. The task is relation reliability, not complete open-vocabulary graph generation.

Violation@K is verifier-derived, and some geometry primitives overlap with the constructed compatibility target. Algebra and counterfactual controls test the learned factor but do not replace an independent physical-validity reference. All three predictors share 3DSSG/3RScan geometry and ontology. Future work should validate the metric independently, improve contact evidence, test cross-dataset transfer, and measure downstream decision benefits.

Conclusion

RelCompat3D isolates source confidence from predicate-conditioned object-pair geometry. A linked-counterfactual objective and exact relation-algebra projection turn that separation into a testable compatibility mechanism. Across three predictors on a fixed 3DSSG target, the structured product improves the aggregate $K = 100$ recall–violation operating point on all three predictors and exhibits predictor-dependent trade-offs at smaller budgets. RelCompat3D contributes a predictor-agnostic reliability framework with explicit family and single-dataset scope limits.

Table 2: **Ablations and counterfactual controls.** R is Recall and V is Violation (lower is better). All rows use the same candidates and training-only model. Label-fixed swap applies only to proximity and vertical relations.

Source	Condition	R50	V50	R100	V100
Open3DSG	RelCompat3D product	.4698	.0286	.6055	.0339
Open3DSG	Wrong predicate	.4507	.2486	.5753	.2315
Open3DSG	Wrong-pair G	.1848	.0814	.2885	.0891
Open3DSG	Shuffled G	.2281	.1418	.3326	.1418
Open3DSG	Label-fixed swap	.4532	.2498	.5775	.2325
Open3DSG	Distance only	.3454	.0846	.5038	.1071
Open3DSG	No source score	.3432	.0186	.5798	.0321
VL-SAT	RelCompat3D product	.9293	.0203	.9688	.0325
VL-SAT	Wrong predicate	.9063	.0391	.9507	.0745
VL-SAT	Wrong-pair G	.8369	.0255	.9121	.0525
VL-SAT	Shuffled G	.8338	.0308	.9089	.0615
VL-SAT	Label-fixed swap	.9068	.0391	.9509	.0745
VL-SAT	Distance only	.3746	.0724	.5554	.0981
VL-SAT	No source score	.3072	.0140	.6186	.0193
SGFN	RelCompat3D product	.7706	.0256	.9418	.0372
SGFN	Wrong predicate	.8097	.0692	.9182	.1120
SGFN	Wrong-pair G	.6619	.0403	.8429	.0723
SGFN	Shuffled G	.6845	.0499	.8338	.0861
SGFN	Label-fixed swap	.8089	.0691	.9187	.1120
SGFN	Distance only	.3746	.0724	.5554	.0981
SGFN	No source score	.3072	.0140	.6186	.0193

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